Linear Algebra

Week 2

Contents

1 Fields, continued: congruence 2

2 Finite fields 3

3 Complex numbers 6

4 Polar representation of complex numbers 8

5 Conjugation and further properties of complex numbers 12

$\mathbb{N} = \{1, 2, 3, \dots\}$ denotes the positive integers.

1 Fields, continued: congruence

We write $\times$ for multiplication of scalars (field elements); juxtaposition such as ab denotes the same product. In integer calculations, $\times$ has its usual multiplication meaning.

1.1 Divisibility and equal remainders

We write $n \mid m$ if m is an integer multiple of n: there is a $k \in \mathbb{Z}$ such that $m = kn$. For example, $4 \mid 20$, whereas $4 \nmid 18$.

Definition 1.1 (Congruence)**.** Let $n \in \mathbb{N}$ and let $a, b \in \mathbb{Z}$. We say that a and b are *congruent modulo* n, and write

$$a \equiv b \pmod{n},$$

if $n \mid (a - b)$.

In ordinary language, a and b have the same remainder after division by n. Congruence is an equality of remainders, not necessarily an equality of integers.

Example 1.1 (Reading a congruence)**.** Modulo 5,

$$17 \equiv 2 \pmod{5}, \qquad 27 \equiv 17 \pmod{5}, \qquad -3 \equiv 2 \pmod{5}.$$

Indeed, $17 - 2 = 15$, $27 - 17 = 10$, and $-3 - 2 = -5$ are all multiples of 5. Modulo 2, two integers are congruent exactly when they have the same parity.

For any positive n, each integer has exactly one representative among $0, 1, \dots, n - 1$. For instance,

$$58 \equiv 2 \pmod{7},$$

and no other number in $\{0, 1, \dots, 6\}$ is congruent to 58 modulo 7.

Theorem 1.2 (Congruence is an equivalence relation)**.** Fix $n \in \mathbb{N}$. Congruence modulo n has the following properties.

1. Reflexivity: $a \equiv a \pmod{n}$ for every $a \in \mathbb{Z}$.
2. Symmetry: if $a \equiv b \pmod{n}$, then $b \equiv a \pmod{n}$.
3. Transitivity: if $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$.

A relation with these three properties is called an *equivalence relation*.

Proof. Reflexivity follows from $a - a = 0 = n \times 0$. For symmetry, if $a - b = nk$, then $b - a = n(-k)$. For transitivity, write $a - b = nk$ and $b - c = n\ell$. Adding gives

$$a - c = (a - b) + (b - c) = n(k + \ell).$$

In each case the relevant difference is divisible by n. □

Example 1.2. Since $9 \equiv 23 \pmod{7}$ and $23 \equiv 44 \pmod{7}$, transitivity gives $9 \equiv 44 \pmod{7}$. Directly, $44 - 9 = 35 = 5 \times 7$.

1.2 Arithmetic respects congruence

Example 1.3. We have $9 \equiv 2 \pmod 7$ and $12 \equiv 5 \pmod 7$. Their sums satisfy

$$9 + 12 = 21 \equiv 0 \pmod 7, \qquad 2 + 5 = 7 \equiv 0 \pmod 7.$$

Their products also have equal remainders:

$$9 \times 12 = 108 \equiv 3 \pmod 7, \qquad 2 \times 5 = 10 \equiv 3 \pmod 7.$$

The next theorem explains why this always works.

Theorem 1.3 (Sums, differences, and products)**.** If $a \equiv b \pmod n$ and $c \equiv d \pmod n$, then

$$a + c \equiv b + d \pmod n, \qquad a - c \equiv b - d \pmod n,$$

and

$$ac \equiv bd \pmod n.$$

Proof. Write $a - b = nk$ and $c - d = n\ell$. Then

$$(a \pm c) - (b \pm d) = n(k \pm \ell).$$

For the product, arrange the difference as

$$ac - bd = (a - b)c + b(c - d) = n(kc + b\ell).$$

Thus every required difference is a multiple of n. □

Example 1.4 (Large numbers without large calculations)**.** 1. Since $10 \equiv 1 \pmod 3$, we have

$$2718 = 2 \times 1000 + 7 \times 100 + 1 \times 10 + 8 \equiv 2 + 7 + 1 + 8 = 18 \equiv 0 \pmod 3.$$

So 2718 is divisible by 3.

2. To find the last digit of 3^{100}, observe that

$$3^4 = 81 \equiv 1 \pmod{10}.$$

Therefore $3^{100} = (3^4)^{25} \equiv 1^{25} = 1 \pmod{10}$, so its last digit is 1.

3. Since $365 \equiv 1 \pmod 7$, the weekday advances by one when the interval between two dates is 365 days. If the interval is 366 days, it advances by two.

4. Since $10^j \equiv 1 \pmod 9$ for every nonnegative integer j, a decimal integer has the same remainder modulo 9 as the sum of its digits. For example,

$$5832 \equiv 5 + 8 + 3 + 2 = 18 \equiv 0 \pmod 9.$$

This is the familiar divisibility test for 9.

Example 1.5 (A rotating weekly schedule)**.** A study group meets every 10 days. If one meeting is on a Monday, the next is $10 \equiv 3 \pmod 7$ weekdays later, hence on Thursday. After seven intervals the total shift is

$$7 \times 10 = 70 \equiv 0 \pmod 7,$$

so the meeting is on a Monday again. Only the remainder of the elapsed time matters for the weekday.

2 Finite fields

2.1 The residue set and its operations

Definition 2.1 (The set $\mathbb{Z}_n$)**.** For an integer $n \geq 2$, define

$$\mathbb{Z}_n = \{0, 1, \ldots, n-1\}.$$

We use these integers as the representatives of the possible remainders modulo n.

For example, $\mathbb{Z}_2 = \{0, 1\}$, $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$, and $\mathbb{Z}_8 = \{0, 1, \ldots, 7\}$.

Theorem 2.2 (Existence and uniqueness of a remainder)**.** For every $a \in \mathbb{Z}$, there is exactly one $r \in \mathbb{Z}_n$ such that $a \equiv r \pmod{n}$.

Proof. The division algorithm gives $a = qn + r$ with $0 \leq r < n$, so $a \equiv r \pmod{n}$. If $r, s \in \mathbb{Z}_n$ are both congruent to a, then $n \mid (r - s)$. But $-(n-1) \leq r - s \leq n - 1$, and the only multiple of n in this interval is 0. Hence $r = s$. □

Definition 2.3 (Addition and multiplication modulo n)**.** For $a, b \in \mathbb{Z}_n$, define $a \oplus b$ to be the unique element of $\mathbb{Z}_n$ congruent to $a + b$ modulo n. Define $a \otimes b$ to be the unique element of $\mathbb{Z}_n$ congruent to ab modulo n.

When the modulus is clear, we normally use the ordinary symbols $+$ and $\times$ in place of $\oplus$ and $\otimes$. Remember that the final result must be reduced to an element of $\mathbb{Z}_n$.

Example 2.1. In $\mathbb{Z}_8$,

$$6 \oplus 5 = 3, \qquad 6 \otimes 5 = 6,$$

because $11 \equiv 3 \pmod{8}$ and $30 \equiv 6 \pmod{8}$. Also,

$$2 \otimes 4 = 0.$$

Both factors are nonzero, so $\mathbb{Z}_8$ has *zero divisors*. Consequently it is not a field with these operations.

Theorem 2.4 (Which field axioms hold in $\mathbb{Z}_n$?)**.** For every $n \geq 2$, modular addition and multiplication satisfy associativity, commutativity, and distributivity. The additive identity is 0, the multiplicative identity is 1, these identities are distinct, and each element has an additive inverse. The only field axiom that can fail is the existence of a multiplicative inverse for every nonzero element.

Proof. Commutativity follows from integer commutativity and uniqueness of the remainder. For associativity of addition, both

$$(a \oplus b) \oplus c \quad \text{and} \quad a \oplus (b \oplus c)$$

are elements of $\mathbb{Z}_n$ congruent to $a + b + c$, so they are equal. For multiplication, both parenthesizations are congruent to abc, giving associativity in exactly the same way. Distributivity follows because

$$a \otimes (b \oplus c) \equiv a(b + c) = ab + ac \equiv (a \otimes b) \oplus (a \otimes c) \pmod{n},$$

and both sides belong to $\mathbb{Z}_n$. Reduction leaves $a + 0$ and $a \times 1$ equal to a. Finally, the additive inverse of 0 is 0, and for $1 \leq a \leq n - 1$ the additive inverse is $n - a$. □

Example 2.2 (Associativity and inverses)**.** In $\mathbb{Z}_9$, take $a = 7$, $b = 8$, and $c = 6$. Then

$$(7 \oplus 8) \oplus 6 = 6 \oplus 6 = 3,$$

whereas

$$7 \oplus (8 \oplus 6) = 7 \oplus 5 = 3.$$

The two parenthesizations agree.

In $\mathbb{Z}_5$, every nonzero element has an inverse:

$$1^{-1} = 1, \qquad 2^{-1} = 3, \qquad 3^{-1} = 2, \qquad 4^{-1} = 4.$$

Indeed, $2 \times 3 = 6 \equiv 1 \pmod 5$ and $4 \times 4 = 16 \equiv 1 \pmod 5$. Thus $\mathbb{Z}_5$ is a field.

In $\mathbb{Z}_8$, the products of 2 with all the elements are

$$0, 2, 4, 6, 0, 2, 4, 6.$$

None is 1, so 2 has no multiplicative inverse there. As a further example, $3 \times 3 = 0$ in $\mathbb{Z}_9$, so $\mathbb{Z}_9$ is not a field either.

Remark 2.5. A multiplicative inverse, when it exists, is unique. If $ab = 1$ and $ac = 1$ in a field, then

$$b = b \times 1 = b(ac) = (ba)c = 1 \times c = c.$$

Also, associativity is a property to check, not a rule that every operation automatically obeys. For instance, subtraction is not associative: $(8 - 3) - 2 = 3$, while $8 - (3 - 2) = 7$.

2.2 Exactly when is $\mathbb{Z}_n$ a field?

Theorem 2.6. The set $\mathbb{Z}_n$, with addition and multiplication modulo n, is a field if and only if n is prime.

Proof. Suppose first that n is composite. Write $n = uv$ with $1 < u < n$ and $1 < v < n$. In $\mathbb{Z}_n$, both u and v are nonzero, but $uv = 0$. If u had an inverse, multiplication by it would give $v = 0$, a contradiction. Thus $\mathbb{Z}_n$ cannot be a field.

Conversely, let $n = p$ be prime and fix $a \in \mathbb{Z}_p$ with $a \neq 0$. Consider

$$a \otimes 0,\ a \otimes 1, \ldots,\ a \otimes (p - 1).$$

These p remainders are all different. Indeed, if $a \otimes j = a \otimes k$, then $p \mid a(j - k)$. Since p is prime and $p \nmid a$, the elementary prime-divisibility rule gives $p \mid (j - k)$. Since $0 \leq j, k < p$, this forces $j = k$.

There are exactly p elements in $\mathbb{Z}_p$, so the list contains every element, including 1. Hence $a \otimes b = 1$ for some $b \in \mathbb{Z}_p$. Every nonzero element therefore has a multiplicative inverse, completing the missing field axiom. □

Once we work in a field, the usual notation is available:

$$-a, \qquad a^{-1} = \frac{1}{a} \quad (a \neq 0), \qquad \frac{a}{b} = a \otimes b^{-1} \quad (b \neq 0).$$

These symbols refer to the operations of that field. They need not have their ordinary rational-number values.

Example 2.3 (A fraction in a finite field)**.** In $\mathbb{Z}_7$, $5^{-1} = 3$ because $5 \times 3 = 15 \equiv 1 \pmod 7$. Therefore

$$\frac{4}{5} = 4 \otimes 3 = 5 \quad \text{in } \mathbb{Z}_7.$$

This is a statement inside $\mathbb{Z}_7$, not the real-number equality $4/5 = 5$.

Example 2.4 (Solving an equation in $\mathbb{Z}_7$)**.** Solve

$$4x + 2 = 6 \quad \text{over } \mathbb{Z}_7.$$

The additive inverse of 2 is 5, because $2 + 5 = 0$ in $\mathbb{Z}_7$. Add 5 to both sides:

$$4x = 6 + 5 = 4.$$

The inverse of 4 is 2, because $4 \times 2 = 8 \equiv 1 \pmod 7$. Multiplying by 2 gives

$$x = 2 \times 4 = 1.$$

Check: $4 \times 1 + 2 = 6$, as required.

Example 2.5 (Why cancellation can fail outside a field)**.** In $\mathbb{Z}_8$, the equation $2x = 2$ has both $x = 1$ and $x = 5$ as solutions, because

$$2 \times 1 = 2, \qquad 2 \times 5 = 10 \equiv 2 \pmod 8.$$

Dividing by 2 is not permitted: 2 has no inverse in $\mathbb{Z}_8$. This example explains why the inverse axiom matters when solving equations.

3 Complex numbers

3.1 Extending the real numbers

Different equations motivate different number systems. The equation $x + 6 = 2$ requires a negative integer; $3x = 2$ requires a rational number; and $x^2 = 3$ has real solutions but no rational solutions. Yet $x^2 + 1 = 0$ has no real solution, because a real square cannot be negative.

We introduce a new number i satisfying $i^2 = -1$. The resulting number system will again be a field.

Definition 3.1 (Complex numbers and their arithmetic)**.** The *imaginary unit* i satisfies $i^2 = -1$. A *complex number* is an expression

$$z = a + ib, \qquad a, b \in \mathbb{R}.$$

The set of complex numbers is

$$\mathbb{C} = \{a + ib : a, b \in \mathbb{R}\}.$$

Addition and multiplication are defined by

$$\begin{aligned}(a + ib) + (c + id) &= (a + c) + i(b + d),\\ (a + ib)(c + id) &= (ac - bd) + i(ad + bc).\end{aligned}$$

Two complex numbers are equal exactly when their real coefficients and their coefficients of i are equal.

Addition acts separately on the two coefficients. To multiply, expand as usual and replace i^2 by -1:

$$(a + ib)(c + id) = ac + iad + ibc + i^2bd = (ac - bd) + i(ad + bc).$$

Example 3.1 (First calculations)**.** 1. The numbers i, $-3i$, $5 + 2i$, and $7 = 7 + 0i$ are complex. Every real number is complex, so $\mathbb{R} \subset \mathbb{C}$.

2. $i(i + 4) = i^2 + 4i = -1 + 4i$.
3. $(3 + 2i)(1 - i) = 3 - 3i + 2i - 2i^2 = 5 - i$.
4. $(2 - i)^2 = 4 - 4i + i^2 = 3 - 4i$. Thus $2 - i$ is a square root of $3 - 4i$; so is $-2 + i$.
5. $(5 + 2i)(5 - 2i) = 25 - (2i)^2 = 25 + 4 = 29$.

Theorem 3.2 (Basic algebraic properties)**.** Complex addition is associative and commutative, 0 is its identity, and the additive inverse of $a+ib$ is $-a-ib$. Complex multiplication is associative and commutative and has identity 1.

Proof. The addition statements follow by comparing the two real coefficients. For example,

$$((a+ib)+(c+id))+(e+if) = (a+c+e)+i(b+d+f),$$

and the other parenthesization gives the same expression.

Commutativity of multiplication follows because $ac-bd = ca-db$ and $ad+bc = cb+da$. For associativity, let $z=a+ib$, $w=c+id$, and $u=e+if$. Then

$$\begin{aligned}(zw)u &= \big((ac-bd)e-(ad+bc)f\big)\\ &\qquad +i\big((ac-bd)f+(ad+bc)e\big)\\ &= (ace-bde-adf-bcf)\\ &\qquad +i(acf-bdf+ade+bce),\end{aligned}$$

whereas

$$\begin{aligned}z(wu) &= \big(a(ce-df)-b(cf+de)\big)\\ &\qquad +i\big(a(cf+de)+b(ce-df)\big)\\ &= (ace-adf-bcf-bde)\\ &\qquad +i(acf+ade+bce-bdf).\end{aligned}$$

The real and imaginary coefficients agree. Finally, $(a+ib)1 = a+ib$. □

Every nonzero complex number has a multiplicative inverse.

3.2 Real and imaginary parts; the complex plane

Represent $z=a+ib$ by the point (a,b) in the plane. The horizontal axis is the *real axis*, and the vertical axis is the *imaginary axis*. This coordinate plane is called the *complex plane*, or the Gauss plane.

Definition 3.3. For $z=a+ib$, define

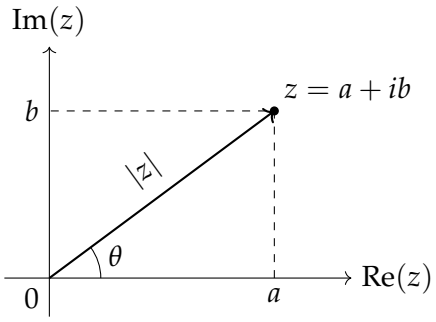
$$\operatorname{Re}(z)=a, \qquad \operatorname{Im}(z)=b.$$

Both $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$ are real numbers. In particular, $\operatorname{Im}(z)$ is b, not ib.

Example 3.2.

z	$\operatorname{Re}(z)$	$\operatorname{Im}(z)$
$-2+5i$	-2	5
$-4i$	0	-4
6	6	0

A number is real exactly when $\operatorname{Im}(z)=0$, in which case $z=\operatorname{Re}(z)$. It is *purely imaginary* when $\operatorname{Re}(z)=0$, so it has the form $z=ib=i\operatorname{Im}(z)$. Under this convention, 0 is both real and purely imaginary.



4 Polar representation of complex numbers

4.1 Modulus and the unit circle

Definition 4.1 (Modulus). For $z = a + ib$, its *modulus*, also called its absolute value, is

$$|z| = \sqrt{a^2 + b^2}.$$

Geometrically, $|z|$ is the distance from (a, b) to the origin. Thus $|z| \geq 0$, with equality exactly when $z = 0$.

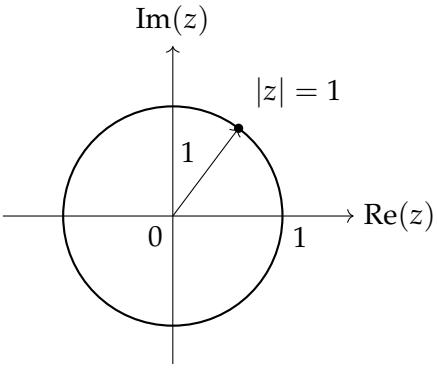
Example 4.1.

$$|-i| = \sqrt{0^2 + (-1)^2} = 1, \qquad |-7| = 7,$$

and

$$|-3 + 4i| = \sqrt{(-3)^2 + 4^2} = \sqrt{25} = 5.$$

The points satisfying $|z| = 1$ form the *unit circle*, whose Cartesian equation is $a^2 + b^2 = 1$. More generally, $|z| = R$ is a circle centered at the origin when $R > 0$. For $R = 0$, it is the single point 0.



4.2 Argument and choosing the correct quadrant

Definition 4.2 (Principal argument). For $z \neq 0$, its *principal argument* $\mathrm{Arg}(z)$ is the angle θ, measured from the positive real axis to the ray from 0 through z, chosen in the range

$$-\pi < \theta \leq \pi.$$

The argument of 0 is undefined.

Counterclockwise angles are positive and clockwise angles are negative. The endpoint convention includes π and excludes $-\pi$.

Example 4.2.

$$\operatorname{Arg}(3i) = \frac{\pi}{2}, \qquad \operatorname{Arg}(-6) = \pi, \qquad \operatorname{Arg}(2-2i) = -\frac{\pi}{4}.$$

For $z = -\sqrt{3} + i$, the point lies in the second quadrant and the angle is $\operatorname{Arg}(z) = 5\pi/6$.

Theorem 4.3 (The tangent relation)**.** Let $z = a + ib \neq 0$ and let $\theta = \operatorname{Arg}(z)$. If $a \neq 0$, then

$$\tan\theta = \frac{b}{a}.$$

If $a = 0$, then $\theta = \pi/2$ or $-\pi/2$, and $\tan\theta$ is undefined.

Proof. The point (a, b) lies on a ray at angle θ. Hence $a = |z|\cos\theta$ and $b = |z|\sin\theta$. If $a \neq 0$, divide the second equality by the first to obtain $b/a = \tan\theta$. If $a = 0$, the ray is vertical. □

Example 4.3 (The tangent does not determine the quadrant)**.** For $z = 2 + 2i$, $b/a = 1$, and the point is in the first quadrant, so $\operatorname{Arg}(z) = \pi/4$. For $w = -2 - 2i$, the ratio is also 1, but the point is in the third quadrant. The principal argument is therefore

$$\operatorname{Arg}(w) = -\frac{3\pi}{4},$$

not $\pi/4$. Always combine the tangent calculation with the location of the point.

4.3 Changing between Cartesian and polar coordinates

Theorem 4.4 (Polar representation)**.** Every nonzero complex number can be written as

$$z = r(\cos\theta + i\sin\theta), \qquad r = |z| > 0, \quad \theta = \operatorname{Arg}(z).$$

For $z = 0$, the same expression is valid with $r = 0$ and any real θ, although $\operatorname{Arg}(0)$ remains undefined.

Proof. If $z = a + ib$ lies at distance r from the origin on the ray of angle θ, elementary trigonometry gives $a = r\cos\theta$ and $b = r\sin\theta$. Substitute these coordinates into $z = a + ib$. □

Example 4.4 (Four polar forms)**.**

$$\begin{aligned} -3i &= 3\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right), \\ -6 &= 6(\cos\pi + i\sin\pi), \\ 2-2i &= 2\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right), \\ -\sqrt{3}+i &= 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right). \end{aligned}$$

Conversely,

$$4\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right) = 4\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) = 2\sqrt{3} + 2i.$$

The conversions can be collected as follows:

polar to Cartesian: $a = r\cos\theta, \quad b = r\sin\theta,$

Cartesian to polar: $r = \sqrt{a^2 + b^2}, \quad \tan\theta = b/a$ if $a \neq 0,$ with the quadrant checked.

4.4 Equality in polar representation

The polar angle need not lie in the principal range. Since sine and cosine are 2π-periodic, adding $2\pi k$ to an angle does not change the represented complex number.

Example 4.5. The same number $-2i$ has the two representations

$$-2i = 2\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right) = 2\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right).$$

Only $-\pi/2$ is its principal argument under our convention.

Theorem 4.5 (Uniqueness up to full turns)**.** If $r \neq 0$ and

$$r(\cos\theta + i\sin\theta) = r(\cos\phi + i\sin\phi),$$

then $\theta = \phi + 2k\pi$ for some $k \in \mathbb{Z}$.

Proof. After dividing by r, compare real and imaginary parts to obtain $\cos\theta = \cos\phi$ and $\sin\theta = \sin\phi$. Thus

$$\cos(\theta - \phi) = \cos\theta\cos\phi + \sin\theta\sin\phi = \cos^2\phi + \sin^2\phi = 1.$$

The real solutions of $\cos t = 1$ are exactly $t = 2k\pi$. Therefore $\theta - \phi = 2k\pi$. □

For genuine polar representations with nonnegative radii, equality also implies equality of the radii, because both radii equal the modulus.

4.5 Multiplication: multiply lengths and add angles

Theorem 4.6 (Multiplication in polar form)**.** If

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1), \qquad z_2 = r_2(\cos\theta_2 + i\sin\theta_2),$$

then

$$z_1 z_2 = r_1 r_2\big(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)\big).$$

Proof. Expanding the product yields

$$\begin{aligned} z_1 z_2 = r_1 r_2\big[&\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 \\ &+ i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)\big]. \end{aligned}$$

The angle-addition formulas identify the real coefficient as $\cos(\theta_1 + \theta_2)$ and the imaginary coefficient as $\sin(\theta_1 + \theta_2)$. □

Thus addition is usually easier in Cartesian form, while multiplication is often easier in polar form.

Example 4.6. Take

$$z_1 = \sqrt{3} + i = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right), \qquad z_2 = 1 + i = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right).$$

Then

$$z_1 z_2 = 2\sqrt{2}\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right).$$

In Cartesian form the same answer is

$$(\sqrt{3} + i)(1 + i) = (\sqrt{3} - 1) + i(\sqrt{3} + 1).$$

Example 4.7 (A quarter-turn is multiplication by i)**.** For any $z = a + ib$,

$$iz = i(a + ib) = -b + ia.$$

The point (a, b) moves to $(-b, a)$, which is a counterclockwise quarter-turn. For instance, $i(2 + i) = -1 + 2i$. Multiplication by i preserves the distance to the origin and adds $\pi/2$ to a polar angle.

Remark 4.7 (A polar angle is not always the principal argument)**.** The sum of two principal arguments may leave $(-\pi, \pi]$. For example,

$$\operatorname{Arg}(-1) + \operatorname{Arg}(-1) = 2\pi,$$

but $\operatorname{Arg}((-1)(-1)) = \operatorname{Arg}(1) = 0$. The angles differ by a full turn. Hence one cannot always write $\operatorname{Arg}(z_1 z_2) = \operatorname{Arg}(z_1) + \operatorname{Arg}(z_2)$ as an equality of real numbers. The corresponding polar representation is nevertheless valid. The same caution applies to differences of arguments in division.

4.6 Reciprocals and division

Theorem 4.8 (Reciprocal and quotient in polar form)**.** Suppose

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1), \qquad z_2 = r_2(\cos\theta_2 + i\sin\theta_2) \neq 0.$$

Then z_2 has the reciprocal

$$z_2^{-1} = \frac{1}{r_2}\big(\cos(-\theta_2) + i\sin(-\theta_2)\big),$$

and

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\big(\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)\big).$$

Proof. Multiply z_2 by the proposed reciprocal. The radii multiply to 1 and the angles add to 0, so the product is $\cos 0 + i\sin 0 = 1$. Multiply z_1 by this reciprocal to obtain the quotient formula. If $z_1 = 0$, choose any polar angle for its zero-radius representation; the quotient remains 0. □

Example 4.8. Using the polar forms above,

$$\frac{\sqrt{3} + i}{1 + i} = \sqrt{2}\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right).$$

The radius is $2/\sqrt{2} = \sqrt{2}$ and the angle is $\pi/6 - \pi/4 = -\pi/12$.

4.7 Powers and de Moivre's formula

Theorem 4.9 (De Moivre's formula)**.** If $z = r(\cos\theta + i\sin\theta)$, then for every positive integer n,

$$z^n = r^n\big(\cos(n\theta) + i\sin(n\theta)\big).$$

If $z \neq 0$, the same formula is valid for every $n \in \mathbb{Z}$, including $n = 0$ and negative integers.

Proof. For $n = 1$ the formula is the given representation. Suppose it holds for $n = k$. Multiplying by z and using the polar multiplication rule gives

$$\begin{aligned} z^{k+1} &= r^k\big(\cos(k\theta) + i\sin(k\theta)\big)\, r(\cos\theta + i\sin\theta) \\ &= r^{k+1}\big(\cos((k+1)\theta) + i\sin((k+1)\theta)\big). \end{aligned}$$

Induction proves the formula for positive integers. For $z \neq 0$, the $n = 0$ case gives $1 = 1$, and applying the positive-power formula to the reciprocal proves the negative cases. □

Example 4.9 (A large power). Since $1 - i = \sqrt{2}\big(\cos(-\pi/4) + i\sin(-\pi/4)\big)$,

$$\begin{aligned}(1-i)^{12} &= (\sqrt{2})^{12}\big(\cos(-3\pi) + i\sin(-3\pi)\big)\\ &= 64(-1+0i) = -64.\end{aligned}$$

The polar form replaces a long expansion by a calculation of one radius and one angle.

5 Conjugation and further properties of complex numbers

5.1 The complex conjugate

Definition 5.1 (Conjugation). For $z = a + ib$, its *complex conjugate* is

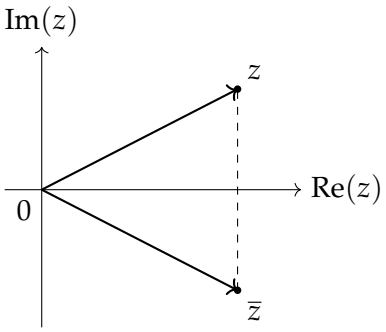
$$\overline{z} = a - ib.$$

Geometrically, conjugation reflects the point across the real axis.

Example 5.1.

$$\overline{4-3i} = 4+3i, \qquad \overline{4+3i} = 4-3i, \qquad \overline{7} = 7, \qquad \overline{-5i} = 5i, \qquad \overline{-9} = -9.$$

Conjugation leaves real numbers fixed and sends a purely imaginary number to its negative.



Theorem 5.2 (Conjugation in polar form). If $z = r(\cos\theta + i\sin\theta)$, then

$$\overline{z} = r\big(\cos(-\theta) + i\sin(-\theta)\big).$$

Proof. Use $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$:

$$\overline{z} = r\cos\theta - ir\sin\theta = r\cos(-\theta) + ir\sin(-\theta).$$

□

For example, the conjugate of $5(\cos(\pi/3) + i\sin(\pi/3))$ is $5(\cos(-\pi/3) + i\sin(-\pi/3))$.

5.2 The basic identities

Theorem 5.3 (Properties of complex numbers). For $z, z_1, z_2 \in \mathbb{C}$, the following identities and inequalities hold.

1. $\mathrm{Re}(z_1 + z_2) = \mathrm{Re}(z_1) + \mathrm{Re}(z_2)$ and $\mathrm{Im}(z_1 + z_2) = \mathrm{Im}(z_1) + \mathrm{Im}(z_2)$.
2. $\overline{\overline{z}} = z$.
3. $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$.

4. $\overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$.
5. If $z_2 \neq 0$, then $\overline{\left(\dfrac{z_1}{z_2}\right)} = \dfrac{\overline{z_1}}{\overline{z_2}}$.
6. $z + \overline{z} = 2\,\mathrm{Re}(z)$.
7. $z - \overline{z} = 2i\,\mathrm{Im}(z)$.
8. $|\,\mathrm{Re}(z)| \leq |z|$ and $|\,\mathrm{Im}(z)| \leq |z|$.
9. $z\overline{z} = |z|^2$.
10. $|\overline{z}| = |z|$.
11. $|z_1 z_2| = |z_1|\,|z_2|$.
12. If $z_2 \neq 0$, then $\left|\dfrac{z_1}{z_2}\right| = \dfrac{|z_1|}{|z_2|}$.
13. The triangle inequality: $|z_1 + z_2| \leq |z_1| + |z_2|$.

Proof. Write $z = a + ib$, $z_1 = a_1 + ib_1$, and $z_2 = a_2 + ib_2$.

1. Real and imaginary parts of a sum. By the definition of addition,

$$z_1 + z_2 = (a_1 + a_2) + i(b_1 + b_2).$$

Thus its real part is $a_1 + a_2$ and its imaginary part is $b_1 + b_2$.

2. Conjugating twice. We have

$$\overline{\overline{z}} = \overline{a - ib} = a + ib = z.$$

3. Conjugate of a sum. On one hand,

$$\overline{z_1 + z_2} = (a_1 + a_2) - i(b_1 + b_2).$$

On the other,

$$\overline{z_1} + \overline{z_2} = (a_1 - ib_1) + (a_2 - ib_2) = (a_1 + a_2) - i(b_1 + b_2).$$

The expressions agree.

4. Conjugate of a product. Multiplication gives

$$z_1 z_2 = (a_1 a_2 - b_1 b_2) + i(a_1 b_2 + b_1 a_2),$$

so

$$\overline{z_1 z_2} = (a_1 a_2 - b_1 b_2) - i(a_1 b_2 + b_1 a_2).$$

Direct multiplication also gives

$$(a_1 - ib_1)(a_2 - ib_2) = (a_1 a_2 - b_1 b_2) - i(a_1 b_2 + b_1 a_2).$$

5. Conjugate of a quotient. Let $q = z_1/z_2$, so $qz_2 = z_1$. Conjugating and using part 4 gives

$$\overline{q}\,\overline{z_2} = \overline{z_1}.$$

Since $z_2 \neq 0$, its conjugate is also nonzero. Dividing by $\overline{z_2}$ gives the claimed formula.

6 and 7. Recovering the real and imaginary parts. Directly,

$$(a + ib) + (a - ib) = 2a = 2\,\mathrm{Re}(z),$$

and

$$(a + ib) - (a - ib) = 2ib = 2i\,\mathrm{Im}(z).$$

In particular, $\mathrm{Re}(z) = (z + \overline{z})/2$ and $\mathrm{Im}(z) = (z - \overline{z})/(2i)$.

8. Bounding the coordinates. Since $b^2 \geq 0$,

$$|\operatorname{Re}(z)| = |a| = \sqrt{a^2} \leq \sqrt{a^2 + b^2} = |z|.$$

The imaginary-part bound follows by interchanging a and b.

9. Product with the conjugate. We have

$$z\overline{z} = (a + ib)(a - ib) = a^2 - (ib)^2 = a^2 + b^2 = |z|^2.$$

10. Conjugation preserves modulus. By definition,

$$|\overline{z}| = \sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2} = |z|.$$

11. Modulus of a product. Using parts 4 and 9 and commutativity,

$$\begin{aligned} |z_1 z_2|^2 &= (z_1 z_2)\overline{z_1 z_2} = z_1 z_2 \overline{z_1}\,\overline{z_2} \\ &= (z_1\overline{z_1})(z_2\overline{z_2}) = |z_1|^2 |z_2|^2. \end{aligned}$$

Taking nonnegative square roots proves the result.

12. Modulus of a quotient. Using parts 5 and 9,

$$\left|\frac{z_1}{z_2}\right|^2 = \frac{z_1}{z_2}\overline{\left(\frac{z_1}{z_2}\right)} = \frac{z_1\overline{z_1}}{z_2\overline{z_2}} = \frac{|z_1|^2}{|z_2|^2}.$$

The denominator is positive, so taking nonnegative square roots is valid.

13. The triangle inequality. Expand the squared modulus:

$$\begin{aligned} |z_1 + z_2|^2 &= (z_1 + z_2)(\overline{z_1} + \overline{z_2}) \\ &= |z_1|^2 + |z_2|^2 + z_1\overline{z_2} + z_2\overline{z_1} \\ &= |z_1|^2 + |z_2|^2 + 2\operatorname{Re}(z_1\overline{z_2}). \end{aligned}$$

By parts 8, 10, and 11,

$$\operatorname{Re}(z_1\overline{z_2}) \leq |z_1\overline{z_2}| = |z_1|\,|z_2|.$$

Consequently,

$$|z_1 + z_2|^2 \leq |z_1|^2 + |z_2|^2 + 2|z_1|\,|z_2| = (|z_1| + |z_2|)^2.$$

Taking nonnegative square roots gives the required inequality. □

Corollary 5.4 (A Cartesian formula for division)**.** For $z \neq 0$,

$$\frac{1}{z} = \frac{\overline{z}}{|z|^2}.$$

More generally, if $z_2 \neq 0$, then

$$\frac{z_1}{z_2} = \frac{z_1\overline{z_2}}{|z_2|^2}.$$

Proof. The identity $z\overline{z} = |z|^2$ and the positivity of $|z|^2$ for $z \neq 0$ give

$$z\left(\frac{\overline{z}}{|z|^2}\right) = 1.$$

Multiplying this reciprocal by z_1 proves the quotient formula. □

Example 5.2 (Division using the conjugate)**.** To write $(3+2i)/(1-2i)$ in Cartesian form, multiply numerator and denominator by $1+2i$:

$$\begin{aligned}\frac{3+2i}{1-2i} &= \frac{(3+2i)(1+2i)}{(1-2i)(1+2i)} \\ &= \frac{3+6i+2i+4i^2}{1+4} = \frac{-1+8i}{5} = -\frac{1}{5}+\frac{8}{5}i.\end{aligned}$$

Check by multiplying back:

$$\left(-\frac{1}{5}+\frac{8}{5}i\right)(1-2i) = \left(-\frac{1}{5}+\frac{16}{5}\right)+i\left(\frac{2}{5}+\frac{8}{5}\right) = 3+2i.$$

Example 5.3 (Distances between two complex numbers)**.** The distance between the points representing $z_1 = 1+2i$ and $z_2 = 4-2i$ is

$$|z_2 - z_1| = |3-4i| = 5.$$

The triangle inequality also bounds the distance from the origin to their sum:

$$|z_1+z_2| = |5| = 5 \le \sqrt{5}+\sqrt{20} = 3\sqrt{5}.$$

This is the usual geometric statement that a straight route is no longer than a route with an intermediate stop.

5.3 The complex numbers form a field

Theorem 5.5. With the addition and multiplication defined above, $\mathbb{C}$ is a field.

Proof. Closure follows from the formulas for the operations. We have proved associativity and commutativity, identified the distinct identities 0 and 1, and given an additive inverse for every number. Every nonzero number has a multiplicative inverse by the polar formula, or equivalently by the conjugate formula.

It remains to check distributivity. For $z = a+ib$, $w = c+id$, and $u = e+if$,

$$\begin{aligned}z(w+u) &= \big(a(c+e)-b(d+f)\big)+i\big(a(d+f)+b(c+e)\big) \\ &= \big((ac-bd)+(ae-bf)\big)+i\big((ad+bc)+(af+be)\big) \\ &= zw+zu.\end{aligned}$$

The other distributive identity follows from commutativity. Thus every field axiom holds. □